



From Experiments to Excellence: The Evolution of Rocket Reliability (1957–2022)

4630

Total Launch Attempts

4555

Unique Missions

66.67%

FirstYearSuccessRate

96.77%

LastYearSuccessRate

89.89%

Overall Success Rate

30.11%

Overall Reliability Improvement

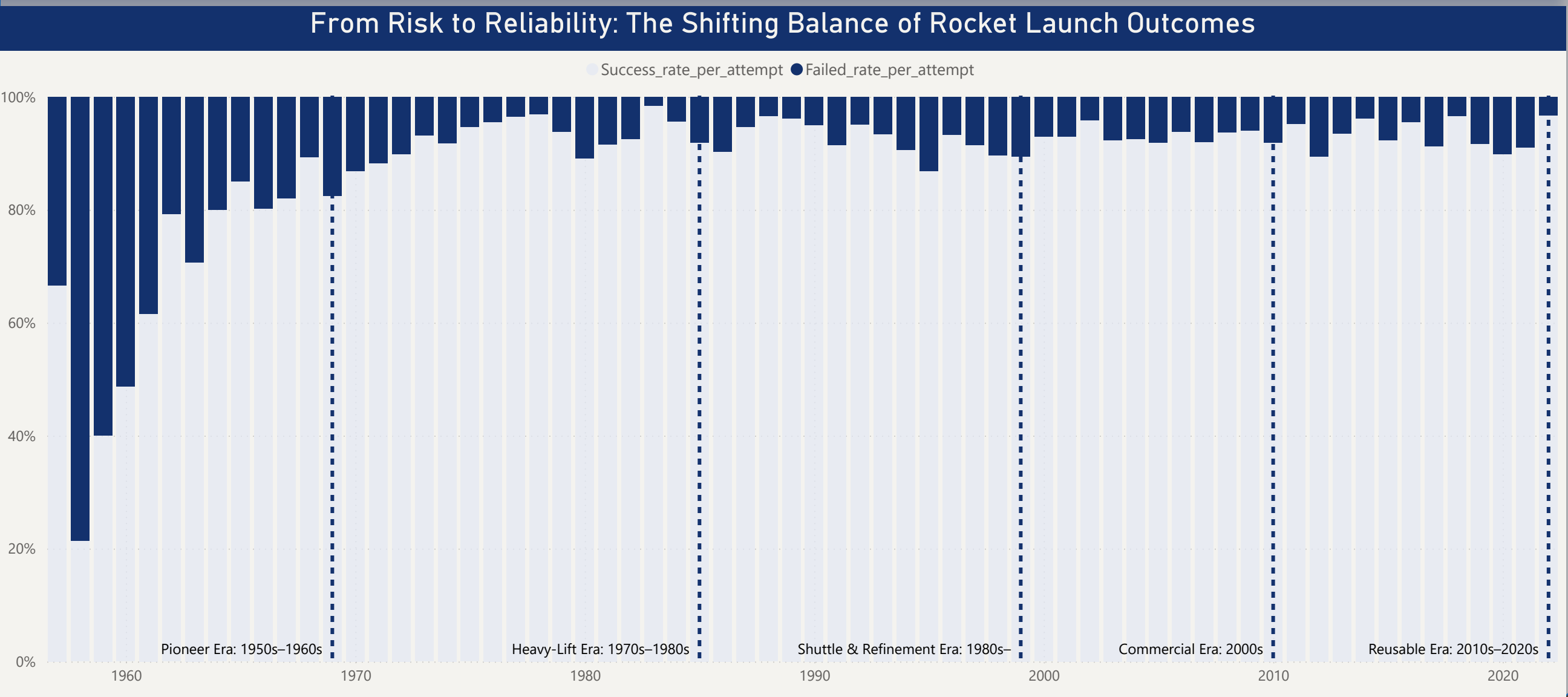
1957

2022

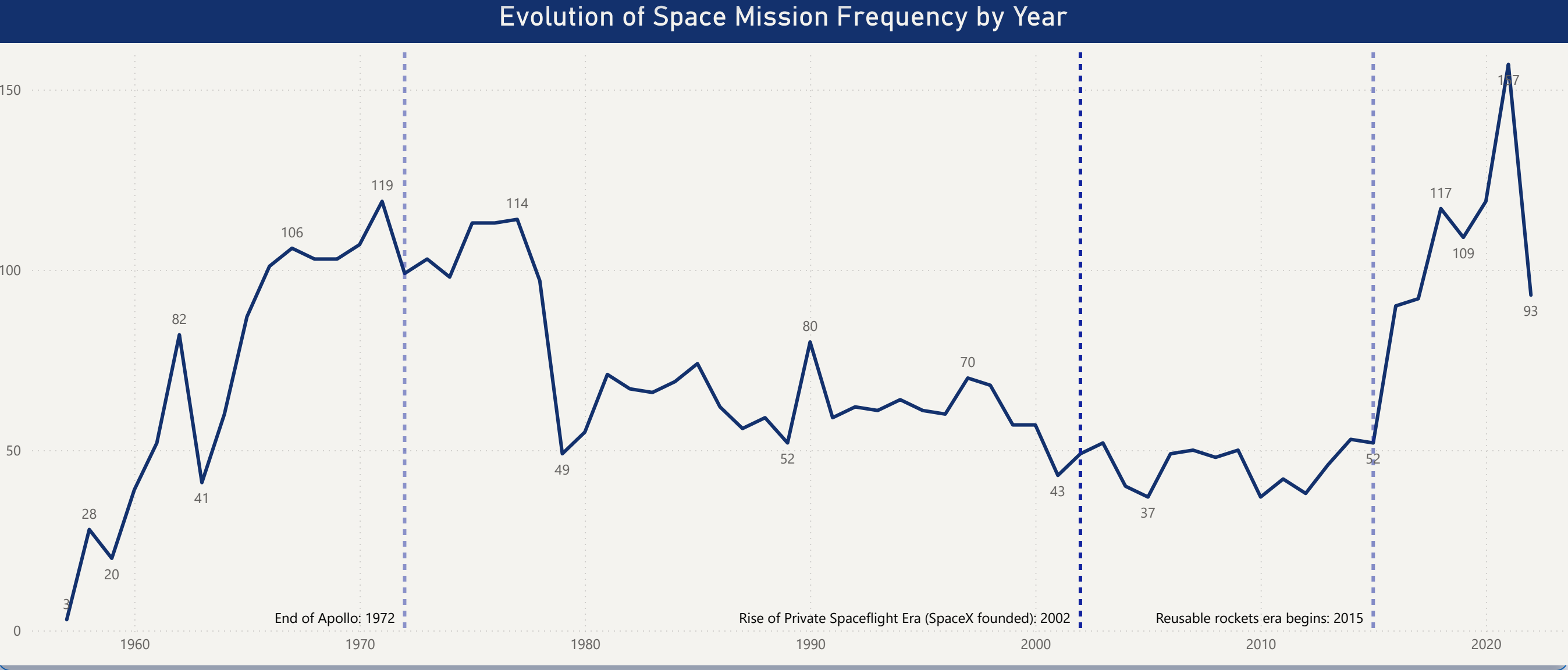


A Timeline of Rocket Performance and Activity

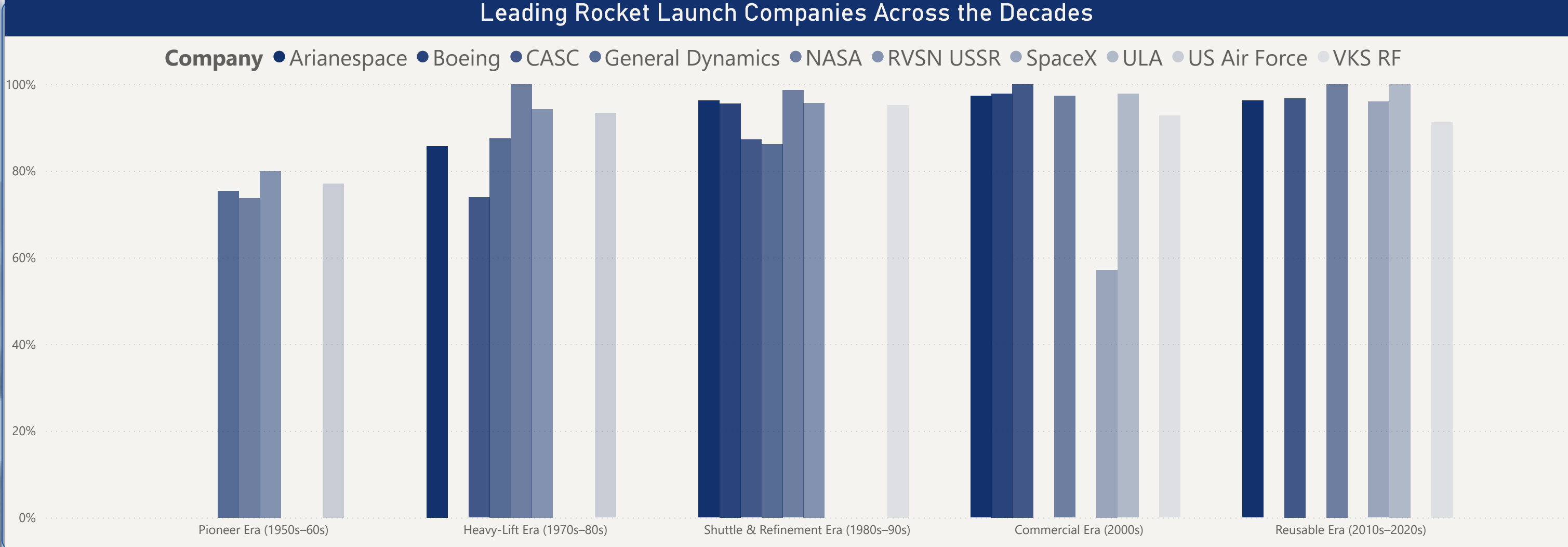
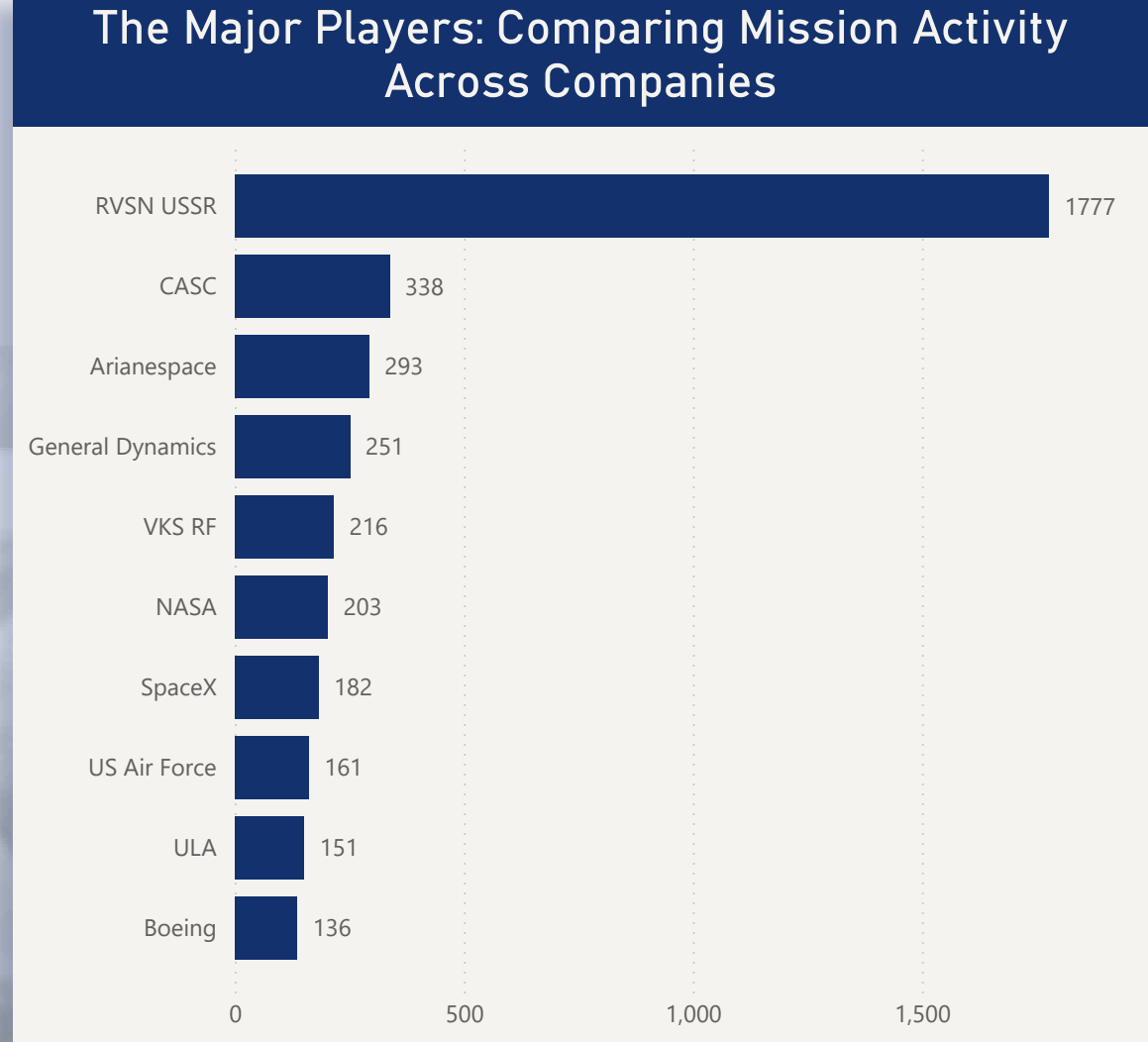
Rocket success climbed from 66.67% in 1957 to 96.77% in 2022, reflecting decades of innovation and engineering progress.



Space activity declined after Apollo, but renewed private and global efforts boosted it from 2015 onward. The 2021 jump reflects delayed launches catching up.



From Legacy to Leadership



2022: The Peak of Modern Rocket Reliability

By 2022 the industry had shifted from one-off experimental flights to engineered repeatability:

- **Reliability:** ~97% mission success.
- **Drivers:** reusable rockets, standardized manufacturing, better telemetry and automated preflight checks.
- **Impact:** fewer mission failures, reduced launch cost, and broader commercial access to orbit.